Learning Money by Living It

A pixel-art life simulation that teaches financial literacy by letting students live it

Team No.: 97

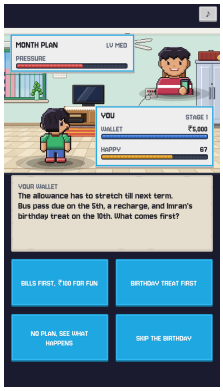
Final report

1. Problem and Our Answer

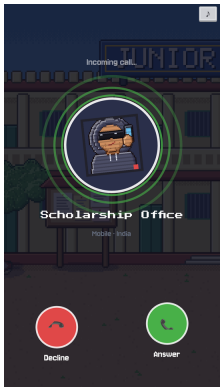
The brief. Design a digital experience that Grade 11-12 (Intermediate) students from underserved communities in Hyderabad *want* to use, and that teaches four pillars: *plan and budget*, *save and manage*, *understand investing*, *stay financially safe*. It starts with about 1,500 students delivered through NGOs and must scale. It is judged on seven measures (simple, relatable, persona-led, language-ready, game-based, intuitive UI, NGO-integrated) under one guardrail: *educate, do not recommend investments; simulations are assumptions, not promises*.

Our reading. Financial literacy here means sensible decisions with one's *own* money: pocket money, a scholarship landing in a first bank account, UPI on a shared phone, a first salary. It is not finance as a subject and not stock trading. The two easy answers both miss: a quiz app with coins is a textbook with confetti, and a trading simulator breaks the guardrail and is far from a 16-year-old's life.

Our answer. A browser game in which the student lives a compressed life (Intermediate student, first salary, earning more, then a look back at 60) and every money decision is a Pokémon-style **money battle** at a Hyderabad place they know. Two claims hold throughout: **the game teaches the concepts while being relatable to the target demographic**, and **the teaching is done through short abstract animations** that play *after* the choice and its consequence, never as a lecture first.



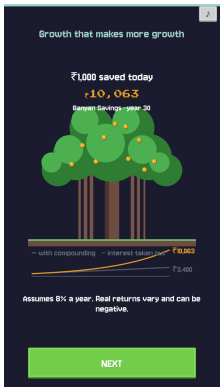
Money battle



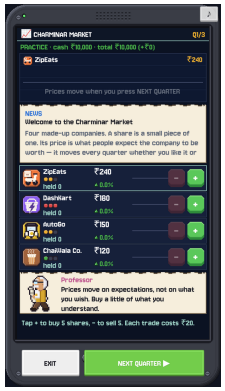
A scam call rings



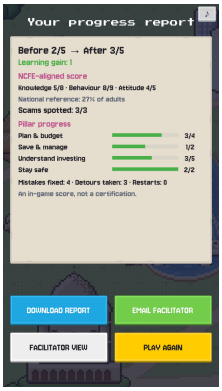
The match you saved for



Compounding as a banyan



Charminar Market



End report

2. Solution

Setup: student, or NGO facilitator with batch code → Avatar, nickname, 4 life questions → persona → 5-question quiz: *before* score + difficulty → Stage 1: student (4 months) → Stage 2: first salary (3 months) → Stage 3: earning more (3 months) → Age 60, what if, *after* quiz, report

The teaching loop. *Situation* (a task from the student's difficulty bucket, framed for their persona) → *choice* (2-4 moves) → *consequence* (wallet and happiness bars move now; some effects are scheduled months later) → *the Professor walks in*: one line of praise for a good choice, or a lesson line plus a 10-20 s abstract animation for a weak one → the concept goes into a *mistake bank* and comes back later in a new disguise until it is answered right. Weak choices send the player down a detour; falling for a scam, running out of money or happiness at zero restarts the task. Good choices can also be *shown* paying off: pick ₹20 chips over a ₹300 biryani and the next day you bat three balls at the turf; order the biryani and the next scene is the match you cannot afford. Money comes from life (allowance, scholarship, wages), not from quests; mini-games are framed as side gigs and lock while a bar is low.

Personas and difficulty are separate. Four picture questions (where money comes from, weekly spend as a range, own bank account, whose phone has UPI) pick one of three personas: *pocket money*, *scholarship* or *earner*. The persona changes only the story: the same scam is an "OTP sent by mistake" on a parent's phone, a fake scholarship officer asking ₹500, or a fake payment screenshot at the shop counter. The quiz score alone picks *simple*, *medium* or *hard*.

End of game. The epilogue at 60 shows what savings and the four made-up assets became, next to a cash box at home. A *what if* screen replays the single biggest mistake from the decision log with the other choice, on the same random seed, side by side (shown only to the student). The after-quiz asks the same five concepts in new situations, giving a learning gain that is not memorisation. The student gets an NCFE-aligned score and a safety score, and a PDF report goes to the NGO facilitator with the guardian in CC.

3. Approach: Why This Design

- **Prices feel like the student's own life.** Stage 1 always gives ₹5,000 in-game a month, and in-game price = real price × ₹5,000 / the student's monthly pocket money (from the indirect weekly-spend band). A ₹200 cricket bat costs ₹200, ₹400 or ₹1,000 in-game for students with ₹5,000, ₹2,500 or ₹1,000 a month: the same *share* of their

money as in real life. Stages 2-3 pay rent, food and travel (55% and 60% of income) every month.

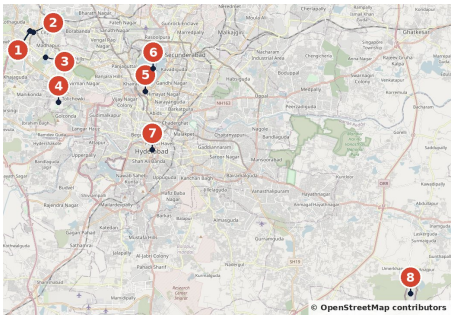
- **Consequence before explanation.** The bars move first, then the Professor explains; the lesson lands on a decision the student owns. Animations are shapes and on-screen text only (a six-operation DSL in JSON), so translating one means swapping strings.
- **Real life is delayed.** 45 delayed or repeating effects: a "no-cost EMI" bites every month, money lent to a friend does not come back, skipped savings meet an emergency with no buffer. Happiness rises with an impulse buy and falls a month later when the student cannot afford the turf match.
- **Local analogies, not imported ones.** Compounding is the Pillalamarri banyan, whose roots become trunks (Hyderabad has no snowballs); diversification is the Charminar vendor who sells umbrellas *and* water bottles.
- **The guardrail is designed in.** Only made-up assets and fictional companies, never a ticker or a "buy"; every projection carries "assumes X% a year; real returns vary and can be negative". The Charminar Market scores *behaviour only* (diversified, held through the crash, refused the hot tip, low churn); money made is shown but never scored.
- **Private by default.** A nickname is the only identifier. The game never asks for OTP, PIN, Aadhaar or a student's phone or email; the facilitator enters report contacts with consent on setup. The sponsor sees aggregates only.

Architecture. One generic battle screen driven by JSON, in four parts with one-way dependencies: *shared* contract types and a zod schema; a *pure engine* in TypeScript (seeded RNG, economy scaler, chapter loop, task selector with pillar rotation and an investing gate, consequence scheduler, mistake bank, investing and market simulations, NCFE scoring, deterministic what-if replay, save and resume); a *content pack* that non-coders and NGOs can edit (tasks per pillar $\times$ difficulty with one framing per persona, detours, events, explainers, places, strings per language); and a *Phaser 3 client* shipped as an offline PWA. A small Node service emails the report and falls back to a local outbox offline; a static dashboard mocks the NGO view.

4. Features Against the Brief

Plan & budget	Needs vs wants, plan the month, compare before buying, borrowing costs extra (incl. "no-cost" EMI): 4 concepts, tasks in every stage.
Save & manage	Save first then spend, emergency buffer, named savings goal; savings feed investing.
Understand investing	Time value, compounding, risk and return, diversification, inflation; four made-up assets with seeded yearly returns; stage jumps of 5, 6 and 25 years make compounding visible.
Stay safe	Fake officials, never share OTP or PIN, too good to be true, instant-loan apps, fake payment screenshots; a separate safety score (scams spotted / faced).
Simple	One decision per screen, 2-4 big buttons, lessons of one line.
Relatable	13 Hyderabad places on 20 original pixel backdrops (Charminar, Durgam Cheruvu, Tank Bund, a kirana, a junior college, a turf ground); prices scaled to the student's own pocket money; TS ePASS scholarships, UPI, chit funds, festival spending.
Persona-led	3 personas $\times$ 24 tasks $=$ 72 framings; 3 difficulty buckets from the quiz; life stages from 16 to 60.
Language-ready	One string file per language: English, Telugu (214/215 UI keys plus content) and Hindi (215/215 UI keys); adding a language means adding a file, with English fallback. The Professor and the characters speak: 422 recorded English voice-over lines, and the phone's own Telugu or Hindi voice elsewhere, with its own on/off switch.
Game-based, tests learning	Before and after quiz (learning gain), NCFE-aligned score: knowledge 0-8, behaviour 0-9, attitude 1-5, *literate* when K$\geq$6, B$\geq$6, A$\geq$3 (27% of Indian adults in NCFE-FLIS 2019); mistakes fixed, detours, restarts.
Interactive & fun	Battles with animated bars; a living city map (day, evening and night, a boat and an auto on the move) with a hub of missions at each landmark; scam calls that ring like a real phone and ring back when declined; a turf match you bat in only if you saved for it; chiptune music; and mini-games: *Kirana Rush* (sort NEED or WANT), *Scam or Legit?* (swipe fake messages) and the *Charminar Market* (four fictional companies, a monsoon, a hot tip and a crash).
Intuitive UI	Tap a landmark to travel; an Info panel (money, commitments, open mistakes, mini-game lock); a replay library of every explainer; hints after two failures; music, sound and voice-over switches on every screen.
NGO-integrated	"Who is setting this up?": the facilitator enters batch code, report email, guardian email and consent, then hands the phone over; PDF report emailed with guardian in CC; facilitator dashboard (mock): before/after, % literate, common mistakes.
Safe to explore	In-game rupees only; detours and restarts instead of game over; the market cannot lose more than was put in.





The city map is Hyderabad. The same eight landmarks, numbered alike, in the game (left) and at their real coordinates (right). Charminar (7) sits due south of Hussain Sagar (6) in both: the same column in the game, 78.47°E in reality, with the Musi river between them. The IT corridor (1-3) is in the west and Ramoji Film City (8) far to the east. For any two landmarks, the game agrees with the city on 47 of 55 east-west and north-south comparisons.
Right: © OpenStreetMap contributors.

5. Results

Engine correctness. 151 unit tests passing in 19 files: engine 77 (RNG, scaler, effects, selector, mistake bank, hints, investing, market, scoring, what-if determinism, save and resume), content 30, mini-games 20, client logic 19, quiz 4, report 1. Typecheck, lint and content validation run on every change.

Content validation. 24 main tasks (12 / 6 / 6 per stage), 72 persona framings, 22 detours, 17 random events, 17 concepts each with an explainer, 45 delayed or repeating effects, 422/422 voice-over lines matched to their clips; every NCFE item (17/17) is evidenced by at least one choice; every cross-reference resolves; a double-blind string check passes.

Scripted players on the real content (20 seeds per difficulty; ranges are across simple, medium and hard). *Good* always picks the best move, *random* picks at random, *worst* picks the worst move but never repeats a fatal one, so it can finish. All 9 balance targets are met.

Policy	Wallet at end	Happiness end (min)	Portfolio at 60	Detours	Restarts	Months broke	Literate	Finished
good	₹0.84-1.02 L	98-99 (66-67)	₹1.36-1.81 L	0	0	0	100%	100%
random	₹19-44 k	32-60 (21-47)	₹0.67-1.26 L	2.7-3.6	0.9-1.4	0.1-0.5	5%	100%
worst	−₹1.77 L to −₹16 k	9-41 (4-39)	0	5.9-7.4	2.9-5.1	0.6-1.6	0%	100%

Good play ends solvent, happy and NCFE-literate; careless play detours about three times and ends with little to spare; the worst play restarts often and goes into debt, yet always finishes. At M2, careless play still ended rich; monthly living costs in Stages 2-3 fixed that.

Play tests. Two unguided team play tests on a phone and in phone mode (02:15 and 06:10) logged 36 issues, each with an owner; 25 were closed by 08:10, including all 12 from the second test. They changed the design: a tighter economy, the Professor as the one teaching character, a labelled happiness number, a heading on every quiz screen, longer mini-game timers (60 s and 45 s), re-worded after-quiz questions, a report PDF that explains why each number matters, the facilitator setup path, and a sound toggle that works everywhere. A request for a stock simulator became the Charminar Market, reshaped to respect the guardrail.

Devices and network (production build in headless Chrome, 360×640, Fast 3G with a 4× CPU throttle, time until the title screen is drawn): **21.0 s** on a first visit (1.1 MB, mostly many small sprite and backdrop files), **1.2 s** on a repeat visit and **1.9 s** for an offline reload through the service worker. The first visit got slower since M2 (5.1 s) as the map, characters and backdrops landed; after one load on Wi-Fi the game runs offline, so a facilitator can prepare phones before a session. **Report pipeline:** a seeded game produces the PDF report, which reaches a sandbox inbox with the guardian in CC; with no network it is saved locally and offered as a download.

Limitations and next steps. Learning gain with real students is not yet measured: the next step is a pilot with one NGO batch, with the dashboard live. Prices and situations are team estimates and need a 10-student survey; Telugu and Hindi need a native-speaker review and their own recorded voice-over (Bhashini for more languages); the first visit on a slow network needs packed sprite atlases and a smaller first download.